Answer key — innate immunity quiz

35 Type A · 15 KPrim · 3 SCT. KPrim: 4/4 = 1 pt, 3/4 = 0.5 pt, ≤2/4 = 0. Sources: Abbas 11e, Janeway 8e, WAO/EAACI HAE guideline 2017.

M1-A1 Type A **Key: D**

A 19-year-old man with cystic fibrosis has repeated lower-airway bacterial infections. Sputum cultures grow bacteria that colonize the airway surface. Blood cultures stay negative and there is no evidence of tissue invasion. Which failure of innate defense best explains this pattern?

A Loss of tight junctions between airway epithelial cells

Tight junctions block passage of microbes between cells. Their loss would favor invasion, but these bacteria do not cross the epithelium. Abbas p.80; Janeway p.44

B Absent alpha-defensin production by Paneth cells

Paneth cells sit in the crypts of the small bowel. They are not the airway defense that fails in cystic fibrosis. Abbas p.80

C Deficient histatin secretion by the salivary glands

Histatins come from the parotid, sublingual and submandibular glands and act mainly against fungi such as Candida. Janeway p.46

D ✔ Impaired mucus clearance by ciliary action

In cystic fibrosis, thick mucus or impaired ciliary movement causes lung infections by bacteria that colonize the epithelial surface but do not cross it. Janeway p.44; Abbas p.80

E Lack of intraepithelial T lymphocytes with limited TCR diversity

Intraepithelial T cells are a normal barrier component. Cystic fibrosis is the textbook example of failed mucus flow, not of absent intraepithelial lymphocytes. Abbas p.80; Janeway p.44

M1-A2 Type A **Key: B**

A 35-year-old man is admitted with community-acquired pneumonia. On day 2 his temperature is 39.2 °C, his blood neutrophil count is raised, and his serum C-reactive protein (CRP) is many times above the upper reference limit. Which mechanism best explains the raised CRP?

A Plasma cells secrete CRP as an early, antigen-specific antibody

CRP is an acute-phase protein from the liver. Acute-phase proteins have no structural diversity and are induced nonspecifically, unlike antibodies. Janeway p.110; Abbas p.97

B ✔ Hepatocytes increase CRP synthesis in response to IL-6, IL-1 and TNF

IL-6, IL-1 and TNF from phagocytes and DCs increase CRP synthesis by liver cells; plasma CRP can rise up to 1000-fold in infection. Abbas p.88; Abbas p.97

C Neutrophils release preformed CRP stored in their granules

The long pentraxin PTX3, not CRP, is stored in neutrophil granules. CRP is made by hepatocytes. Abbas p.87; Abbas p.88

D Macrophage IL-10 drives hepatic CRP production

IL-10 is a negative-feedback cytokine that inhibits macrophages and DCs. It does not drive the acute-phase response. Abbas p.101

E The hypothalamus secretes CRP as part of the febrile response

TNF and IL-1 act on the hypothalamus to cause fever. CRP comes from hepatocytes, not from the hypothalamus. Abbas p.96; Abbas p.97

M1-A3 Type A **Key: E**

A vaccine team tests a purified recombinant protein antigen in healthy volunteers. Alone, the protein induces only weak T cell–dependent antibody responses. When the team adds an aluminium salt, the responses become strong. What is the main reason for this effect?

A Aluminium increases the diversity of B-cell receptor genes
Receptor diversity comes from somatic recombination of gene segments in lymphocytes, not from adjuvants. Abbas p.4

B Aluminium cross-links the T-cell receptor and so replaces signal 1
Signal 1 is antigen recognition. Adjuvants act on the innate side and increase MHC molecules and costimulators on DCs. Abbas p.100; Abbas p.101

C Aluminium blocks IL-10 negative feedback in macrophages
IL-10 is a negative-feedback regulator of macrophages and DCs. The sources describe adjuvant action as innate stimulation, not IL-10 blockade. Abbas p.101

D Aluminium induces memory in innate cells, so later responses are faster
Innate memory is limited, and innate responses to repeated exposure are usually the same as the first. Abbas p.4

E ✔ Aluminium activates innate responses that make DCs express costimulators and cytokines (signal 2)
Aluminium salts may act by inflammasome activation. Adjuvants make DCs express more MHC, costimulators (signal 2) and cytokines, and migrate to lymph nodes. Abbas p.101

M1-K1 KPrim **Key: + / – / + / –**

Innate recognition and the epithelial barrier. Which of the following statements is/are correct?

A (+) Innate immune receptors are germline-encoded and distributed nonclonally: all cells of one lineage carry identical receptors.
Correct. Innate receptors are germline-encoded and nonclonal; adaptive receptors are clonal. Abbas p.63; Abbas p.64

B (–) Lysozyme is more effective against Gram-negative than Gram-positive bacteria because it cleaves lipopolysaccharide.
Incorrect. Lysozyme cleaves peptidoglycan and is more effective against Gram-positive bacteria. In Gram-negative bacteria, LPS covers the peptidoglycan. Janeway p.45

C (+) Paneth cells in the crypts of the small bowel are a major source of alpha-defensins.
Correct. Paneth cells are a major producer of alpha-defensins. Abbas p.80

D (–) Human neutrophils produce functional theta-defensins as their main antimicrobial peptide.
Incorrect. The single human theta-defensin gene is inactivated by a mutation. Janeway p.46

M1-K2 KPrim **Key: + / + / – / +**

Systemic effects of innate cytokines. Which of the following statements is/are correct?

A (+) Endogenous pyrogens induce cyclooxygenase-2 to make prostaglandin E2, which acts on the hypothalamus; aspirin reduces fever by blocking prostaglandin synthesis.
Correct. TNF, IL-1 and IL-6 induce COX-2 and PGE2; prostaglandin synthesis inhibitors such as aspirin reduce fever. Janeway p.109; Abbas p.97

B (+) Acute-phase proteins have broad specificity for microbial patterns but no structural diversity.
Correct. Acute-phase proteins have broad specificity for PAMPs and antibody-like functions, but no structural diversity; they are induced nonspecifically. Janeway p.109; Janeway p.110

C (–) Anti-TNF antibodies have shown a clear survival benefit in clinical trials in patients with sepsis.
Incorrect. Anti-TNF antibodies and soluble TNF receptors have not shown benefit in patients with sepsis. Abbas p.97

D (+) TNF blockade can reactivate tuberculosis, which shows the role of TNF in keeping infection local.
Correct. TNF blockade can reactivate tuberculosis in apparently well patients with previous infection. Janeway p.109; Abbas p.94

M2-A1 Type A **Key: B**

A 46-year-old woman develops generalised urticaria and hypotension within minutes of an intravenous antibiotic given at induction of anaesthesia. She has never received this drug before. Specific IgE to the drug is negative. Which mast cell receptor most likely mediated this IgE-independent reaction?

A FcεRII (CD23)
CD23 is a low-affinity IgE receptor whose biologic role is not known; it does not explain an IgE-independent drug reaction. Abbas p.467

B ✔ MRGPRX2
MRGPRX2 on skin mast cells binds several antibiotics and anaesthetic drugs and activates mast cells without IgE; it likely mediates some allergy-like drug reactions. Abbas p.470

C FcγRIIB
FcγRIIB is an inhibitory (ITIM) receptor that co-aggregates with FcεRI and inhibits mast cell signalling; it does not trigger degranulation. Abbas p.470

D CD16 (FcγRIIIA)
CD16 is the low-affinity IgG receptor on NK cells that mediates ADCC; it is not the drug receptor on mast cells. Abbas p.84

E NKG2D
NKG2D is an activating NK cell receptor for stress ligands (MIC-A, MIC-B, ULBP), not a mast cell drug receptor. Abbas p.83

M2-A2 Type A **Key: D**

A 52-year-old man has had three episodes of unexplained anaphylaxis. Between episodes his baseline serum tryptase stays raised. A bone marrow biopsy shows clusters of mast cells with abnormal morphology. A mutation in which gene is most likely?

A GATA2
GATA2 loss-of-function causes loss of NK cells, monocytes, DCs and B cells, not a clonal mast cell disorder. Abbas p.488

B CD18 (β2 integrin β chain)
CD18 mutations cause leukocyte adhesion deficiency type 1 with bacterial infections, not mast cell expansion. Abbas p.487

C FcεRI α chain
FcεRI is the high-affinity IgE receptor on mast cells and basophils; it is not the gene linked to systemic mastocytosis in the sources. Abbas p.466

D ✔ c-KIT
Systemic mastocytosis is a clonal mast cell disorder associated with c-KIT mutations; the mast cells release more mediators, including tryptase. Abbas p.470-471

E LYST
LYST mutations cause Chédiak-Higashi syndrome with defective cytotoxic granules, not systemic mastocytosis. Abbas p.488

M2-A3 Type A **Key: A**

In a laboratory assay, a tumour cell line that has lost all surface MHC class I is mixed with blood lymphocytes from a healthy donor who was never exposed to the tumour. CD56-positive, CD3-negative cells lyse the tumour cells within hours. No antibody is present. Which mechanism best explains the killing?

A ✔ Loss of inhibitory signals from KIRs and CD94/NKG2A ('missing self')
The cells are NK cells. Inhibitory KIRs and CD94/NKG2A recognise MHC-I; when MHC-I is lost, inhibition stops and activating signals trigger killing. Abbas p.85

B T-cell receptor recognition of tumour peptide on MHC class I
CD3-negative cells have no T-cell receptor, and the target has no MHC-I to present peptide. Abbas p.82

C ADCC through CD16 binding to IgG on the tumour cells
ADCC needs IgG1 or IgG3 bound to the target; no antibody is present. Abbas p.84

D Engagement of the activating receptor CD94/NKG2A by HLA-E
CD94/NKG2A is an inhibitory receptor; it binds HLA-E, which is an MHC-I molecule that the tumour has lost. Abbas p.85

E Release of IL-12 that directly lyses the tumour cells
IL-12 enhances NK cytotoxicity and IFN-γ secretion; it is a cytokine signal, not a killing mechanism. Abbas p.85

M2-A4 Type A **Key: C**

A 16-year-old girl has had recurrent severe herpesvirus infections and extensive warts since childhood. Her blood shows almost no NK cells, and her monocyte, dendritic cell and B-cell counts are also low. Her mother has a similar history. Which is the most likely diagnosis?

A Chédiak-Higashi syndrome
Chédiak-Higashi syndrome is autosomal recessive; NK cell function is impaired by abnormal granules, which does not fit a dominant pattern with low monocytes, DCs and B cells. Abbas p.488

B CD16 (FcγRIIIA) deficiency
CD16 deficiency is a functional NK cell deficiency (autosomal recessive); NK numbers and other lineages are not low. Abbas p.488

C ✔ GATA2 deficiency
Autosomal dominant GATA2 mutations cause loss of NK cells together with low monocytes, DCs and B cells; NK deficiency presents with herpesvirus and papillomavirus infections. Abbas p.488

D Leukocyte adhesion deficiency type 1
LAD-1 (CD18 mutation) causes bacterial and fungal infections from defective leukocyte adhesion, not loss of NK cells. Abbas p.487

E Chronic granulomatous disease
CGD is a defect of phagocyte oxidase with bacterial and fungal infections; it does not remove NK cells, DCs or B cells. Abbas p.487

M2-A5 Type A **Key: E**

A 27-year-old woman with allergic asthma has airway eosinophilia and mucus hypersecretion within hours of allergen exposure. Biopsy studies suggest that the first IL-5 and IL-13 come from tissue-resident cells that lack a T-cell receptor and that respond to IL-33, IL-25 and TSLP from the epithelium. Which cell is this?

A ILC1
ILC1s express T-BET and produce IFN-γ; they respond to IL-12, IL-15 and IL-18. Abbas p.81

B ILC3
ILC3s express RORγt and produce IL-17 and/or IL-22; they respond to IL-1 and IL-23. Abbas p.81

C NK cell
NK cells are cytotoxic and secrete IFN-γ, not IL-5 and IL-13. Abbas p.82

D Th2 cell
Th2 cells make IL-5 and IL-13 but carry a T-cell receptor and arrive later in the tissue than ILC2s. Abbas p.465

E ✔ ILC2
ILC2s (GATA3) lack a TCR, reside in tissue, respond to IL-25, IL-33 and TSLP, and make IL-5 and IL-13; they can start allergic inflammation before Th2 cells arrive. Abbas p.94

M2-A6 Type A **Key: B**

A 38-year-old man with severe allergic asthma produces thick sputum. Microscopy shows many eosinophils and Charcot-Leyden crystals. Which eosinophil protein forms these crystals?

A Major basic protein
Major basic protein is a cationic granule protein toxic to helminths, bacteria and host cells; it does not form Charcot-Leyden crystals. Abbas p.473

B ✔ Galectin-10
Eosinophils make large amounts of galectin-10, which aggregates into Charcot-Leyden crystals at sites of chronic type 2 inflammation such as asthmatic bronchi. Abbas p.473

C Eosinophil peroxidase
Eosinophil peroxidase makes hypochlorous or hypobromous acid; it is not the crystal-forming protein. Abbas p.473

D Tryptase
Tryptase is a mast cell protease and a marker of mast cell activation. Abbas p.471

E Eosinophil cationic protein
Eosinophil cationic protein is a toxic granule protein, but galectin-10 forms the crystals. Abbas p.473

M2-K1 KPrim **Key: + / – / – / +**

Mast cells, basophils and eosinophils in allergic disease. Which of the following statements is/are correct?

A (+) Basophils express high levels of FcεRI and can be triggered when antigen binds IgE on their surface.
Correct. Basophils, like mast cells, have high FcεRI expression, bind IgE and are triggered by antigen binding to the IgE. Abbas p.465; Abbas p.466

B (–) Eosinophils are efficiently activated by allergen cross-linking of IgE bound to their FcεRI.
Incorrect. Eosinophil FcεRI is very low and lacks a signalling chain; there is no convincing evidence of activation through IgE. Abbas p.473

C (–) Mature mast cells circulate in the blood and migrate to tissues during allergic reactions.
Incorrect. Mature mast cells are normally not found in the circulation; progenitors mature in tissues. Abbas p.465

D (+) Tryptase is present in all human mast cells, whereas chymase is present only in some mast cell subsets.
Correct. Tryptase is in all human mast cells; subsets contain tryptase only or tryptase and chymase. Abbas p.465

M2-K2 KPrim **Key: + / – / + / –**

Dendritic cells and NK cells. Which of the following statements is/are correct?

A (+) Plasmacytoid dendritic cells express high levels of TLR7 and TLR9 and are the main producers of type I interferon.
Correct. pDCs have high TLR7 and TLR9 and are the body's major producers of type I IFNs. Abbas p.23

B (–) CD16 on NK cells binds IgE and mediates antibody-dependent cell-mediated cytotoxicity.
Incorrect. CD16 (FcγRIIIA) is a low-affinity receptor for IgG (IgG1 and IgG3), not IgE. Abbas p.84

C (+) The inhibitory receptor CD94/NKG2A recognises HLA-E, which displays peptides from other MHC class I molecules.
Correct. CD94/NKG2A binds HLA-E; HLA-E displays peptides derived from other MHC-I molecules. Abbas p.85

D (–) Follicular dendritic cells are bone marrow-derived cells that present protein antigens to naive T cells.
Incorrect. Follicular dendritic cells are stromal cells, not bone marrow-derived, and do not present protein antigens to T cells. Abbas p.22

M3-A1 Type A **Key: C**

A 2-year-old boy has his second episode of invasive pneumococcal infection (pneumonia with bacteraemia). His parents report that he hardly had a fever during either episode, and the local signs of inflammation were mild. He has had several common viral infections without complications. Which defect best explains this picture?

A Loss-of-function mutation in TLR3
TLR3 loss of function gives recurrent herpes simplex encephalitis, not invasive pyogenic infection. Abbas p.489

B Mutation in the IL-12 receptor β1 chain
IL-12Rβ1 mutations give susceptibility to Salmonella and mycobacteria, not pneumococcal 'cold' infection. Abbas p.93

C ✔ Autosomal recessive IRAK4 deficiency
MyD88/IRAK4 deficiency gives severe invasive bacterial (especially pneumococcal) infections early in life with little fever or inflammation; viral susceptibility is not increased. Abbas p.489; Janeway p.92, 535

D Gain-of-function mutation in NLRP3
NLRP3 gain of function (CAPS) gives excess IL-1 with fever and rash, the opposite of a fever-poor picture. Abbas p.74-75

E Homozygous mutation in UNC93B
UNC93B deficiency impairs endosomal TLRs and type I IFN and predisposes to herpes simplex encephalitis. Abbas p.489

M3-A2 Type A **Key: E**

A 6-year-old girl is admitted with her second episode of herpes simplex virus-1 encephalitis. She has no history of severe bacterial, fungal or other viral infections. Immunoglobulins and lymphocyte subsets are normal. A defect in which pathway is most likely?

A MyD88–IRAK4 signalling to NF-κB
MyD88/IRAK4 defects give invasive pyogenic bacterial infections; IRAK4 deficiency adds no viral susceptibility. Abbas p.489; Janeway p.92

B TLR5 recognition of bacterial flagellin
TLR5 is a surface receptor for flagellin of motile bacteria and has no role in herpes simplex encephalitis in the sources. Abbas p.67

C NOD2 recognition of muramyl dipeptide
NOD2 senses bacterial peptidoglycan (MDP); its variants relate to Crohn's disease and Blau syndrome. Abbas p.72

D Lectin pathway initiation by MBL
Low MBL increases risk of various infections mainly when other immune defects coexist; it does not cause isolated HSE. Abbas p.88

E ✔ Endosomal TLR3–TRIF signalling to type I IFN
Defects in TLR3, UNC93B, TRIF, TRAF3 or TBK1 impair type I IFN downstream of TLR3 and cause herpes simplex encephalitis without broad viral susceptibility. Abbas p.489

M3-A3 Type A **Key: A**

A 34-year-old man has had attacks since childhood of fever, urticarial rash, joint pain and conjunctivitis. The attacks start a few hours after cold exposure. His father has the same attacks. Which mechanism best explains his disease?

A ✔ Gain-of-function NLRP3 mutation with excess inflammasome-generated IL-1β
Dominant NLRP3 (cryopyrin) mutations cause CAPS; in FCAS cold exposure triggers fever, urticarial rash, joint pain and conjunctivitis. Abbas p.74; Janeway p.537

B MEFV mutation with dysregulated pyrin inflammasome
FMF (MEFV/pyrin) presents with fever, serositis and arthritis, without cold-induced urticarial rash. Abbas p.74-75

C TNFRSF1A mutation affecting TNF receptor I
TRAPS gives periodic fever, myalgia and rash; cold triggering is not a feature in the sources. Janeway p.536-537

D Gain-of-function NOD2 mutation with excess NF-κB
NOD2 gain of function causes Blau syndrome with spontaneous inflammation of joints, eyes and skin. Abbas p.72; Janeway p.93

E Partial loss of mevalonate kinase activity
HIDS gives fever from infancy with high serum IgD and lymphadenopathy. Janeway p.537

M3-A4 Type A **Key: D**

A 58-year-old man has a third attack of acute monoarthritis of the first toe. Joint aspirate shows monosodium urate crystals. He does not respond to conventional anti-inflammatory drugs, and the rheumatologist proposes IL-1 blockade. Which innate mechanism explains why IL-1 blockade is effective?

A Urate crystals bind TLR4 and signal through MyD88 alone
TLR4 recognizes LPS (with MD2) and heat shock proteins; the sources link urate crystals to the inflammasome, not to TLR4. Abbas p.67, 74

B Cytosolic DNA from crystals activates cGAS–STING
cGAS–STING senses cytosolic dsDNA and induces type I IFN, not IL-1β. Abbas p.76

C NOD2 in Paneth cells recognizes urate and activates RIP2
NOD2 recognizes muramyl dipeptide from bacterial peptidoglycan and activates NF-κB through RIP2. Abbas p.72

D ✔ Phagocytosed crystals damage lysosomes and activate NLRP3–caspase-1
Urate crystals damage lysosomes and activate the NLRP3 inflammasome; caspase-1 then generates active IL-1β. IL-1 antagonists treat severe resistant gout. Abbas p.72, 74

E Cytosolic urate activates caspase-4 and caspase-5 directly
The noncanonical caspase-4/5 pathway is activated by cytosolic LPS, not by urate. Abbas p.74

M3-A5 Type A **Key: B**

A 45-year-old woman with gram-negative urosepsis has a plasma CRP that rises steeply over 48 hours. Which group of cytokines, produced after TLR4 recognition of LPS, mainly drives this hepatic CRP synthesis?

A IL-12 and IL-18
IL-12 and IL-18 mainly drive IFN-γ production and NK cell function, not acute-phase protein synthesis. Abbas p.93-94

B ✔ IL-6, IL-1 and TNF
IL-6, IL-1 and TNF induce hepatocytes to make acute-phase proteins; CRP can rise up to 1000-fold. Abbas p.88, 97

C IFN-α and IFN-β
Type I IFNs induce an antiviral state and more MHC class I; they are not the drivers of CRP synthesis. Abbas p.90

D IL-33 released as an alarmin
IL-33 is an epithelial alarmin released from damaged cells; it is not the hepatic acute-phase signal. Abbas p.94

E CXCL8 and CCL2
These chemokines recruit neutrophils and monocytes; they do not induce hepatic acute-phase proteins. Abbas p.94

M3-A6 Type A **Key: E**

A 5-month-old girl has recurrent fever, skin lesions due to cutaneous vasculopathy, and tachypnoea. Chest CT shows interstitial lung disease. Infection screening is negative. Which mechanism is most likely?

A Gain-of-function NLRP3 with excess IL-1β
CAPS gives fever, neutrophilic rash, bony lesions and CNS inflammation, not vasculopathy with interstitial lung disease. Abbas p.75

B Loss-of-function TLR3 with low type I IFN
TLR3 loss of function gives herpes simplex encephalitis, not a sterile inflammatory disease. Abbas p.489

C Loss-of-function MyD88 with absent NF-κB signalling
MyD88 deficiency gives invasive pyogenic infections with little inflammation. Abbas p.489

D Gain-of-function NOD2 with excess NF-κB signalling
NOD2 gain of function causes Blau syndrome with inflammation of joints, eyes and skin. Abbas p.72; Janeway p.93

E ✔ Gain-of-function STING with excess type I IFN
STING gain of function causes SAVI, an interferonopathy with cutaneous vasculopathy, fever and interstitial lung disease. Abbas p.75, 77

M3-K1 KPrim **Key: + / + / – / –**

Toll-like receptors. Which of the following statements is/are correct?

A (+) TLR9 sits in endosomes and recognizes unmethylated CpG DNA.
TLR9 is endosomal and binds unmethylated CpG motifs, common in prokaryotic DNA but rare in vertebrate DNA. Abbas p.67, 69

B (+) For LPS recognition, the accessory protein MD2 binds lipid A and the complex engages TLR4.
MD2 binds the lipid A component of LPS and the complex interacts with TLR4; CD14 is also needed for efficient signalling. Abbas p.69

C (–) TLR5 sits in endosomes and recognizes viral single-stranded RNA.
TLR5 is a plasma membrane TLR that binds bacterial flagellin; TLR7 and TLR8 bind ssRNA. Abbas p.67, 69

D (–) TLR3 depends on MyD88 to induce IFN-β.
TLR3 is the only TLR that does not use MyD88; it signals only through TRIF. Abbas p.69

M3-K2 KPrim **Key: + / – / + / +**

NOD-like receptors and inflammasomes. Which of the following statements is/are correct?

A (+) Caspase-1 cleaves gasdermin D, and the N-terminal fragment forms pores in the plasma membrane.
The gasdermin D N-terminal fragment polymerizes into pores through which IL-1β leaves the cell. Abbas p.72

B (–) Neutrophils are the main cell type that dies by pyroptosis.
Pyroptosis affects macrophages and dendritic cells, not neutrophils. Abbas p.74

C (+) NOD2 recognizes muramyl dipeptide, and some NOD2 polymorphisms increase the risk of Crohn's disease.
NOD2 senses MDP from gram-negative and gram-positive peptidoglycan; some variants raise Crohn's risk. Abbas p.72

D (+) In humans, cytosolic LPS can activate caspase-4 or caspase-5 and cause pyroptosis.
This is the noncanonical inflammasome pathway (caspase-11 in rodents). Abbas p.74

M3-K3 KPrim **Key: + / – / + / –**

Soluble PRRs and cytosolic nucleic-acid sensors. Which of the following statements is/are correct?

A (+) CRP binds phosphorylcholine and activates the classical complement pathway by binding C1q.
CRP binds phosphorylcholine; CRP, SAP and PTX3 activate the classical pathway through C1q. Abbas p.87-88

B (–) Ficolins recognize carbohydrates through a C-type lectin domain, like MBL.
Ficolins have a fibrinogen-type recognition domain instead of a C-type lectin domain. Abbas p.89

C (+) RIG-I recognizes viral RNA with a 5′-triphosphate, which capped host mRNA lacks.
RIG-I only recognizes RNA with a 5′-triphosphate; host mRNA carries a 7-methylguanosine cap. Abbas p.75-76

D (–) MAVS is an endoplasmic reticulum adaptor that binds cGAMP.
STING is the ER adaptor that binds cGAMP; MAVS is on the outer mitochondrial membrane and serves RIG-I and MDA5. Abbas p.76

M4-A1 Type A **Key: C**

An 8-month-old boy has recurrent bacterial skin infections and a wound that heals poorly. The infected sites show no pus. His blood leukocyte count is high. Flow cytometry shows very low CD11a, CD11b and CD11c on his neutrophils. Which gene is most likely mutated?

A GDP-fucose transporter
This causes LAD-2: sialyl Lewis X is absent, but β2 integrin (CD11/CD18) expression is not the primary defect. Abbas p.488

B KINDLIN-3
This causes LAD-3: integrins are present but cannot be activated by inside-out signalling. Abbas p.488

C ✔ CD18 (β2 integrin chain)
LAD-1 is caused by CD18 mutations, with absent or reduced β2 integrins (LFA-1, MAC-1, p150,95). Leukocytosis, poor wound healing and no pus fit. Abbas p.488

D LYST
LYST mutations cause Chédiak–Higashi syndrome, with giant lysosomes and partial albinism, not loss of CD11/CD18. Abbas p.488

E gp91 subunit of cytochrome b558
This causes X-linked CGD, a killing defect. Neutrophils reach tissue and integrin expression is normal. Abbas p.486

M4-A2 Type A **Key: E**

A 2-year-old girl has recurrent bacterial infections and a high blood leukocyte count. She also has severe mucosal bleeding that needed blood transfusions. Her platelet count is normal. Her skin, hair and eye pigmentation are normal. Neutrophil CD18 expression by flow cytometry is normal. Which mechanism best explains her disease?

A Absent expression of β2 integrins on leukocytes
CD18 expression is normal, so LAD-1 is unlikely. LAD-1 also does not cause a bleeding disorder. Abbas p.488

B Absence of sialyl Lewis X on leukocytes
LAD-2 blocks rolling and causes developmental abnormalities, not a platelet aggregation defect. Abbas p.488

C Failure of NADPH oxidase to make superoxide
CGD is a killing defect with granulomas and Aspergillus infections; it does not cause bleeding or block adhesion. Abbas p.486-487

D Defective intracellular trafficking of lysosomes
This is CHS (LYST). CHS can cause bleeding from platelet dysfunction, but it also gives partial albinism, silver hair and giant lysosomes. Her normal pigmentation, leukocytosis and normal CD18 favour LAD-3. Abbas p.488; Janeway p.540

E ✔ Defective inside-out activation of integrins
LAD-3 (KINDLIN-3) blocks chemokine-induced integrin activation in leukocytes and platelets. Platelet counts are normal but aggregation fails, so bleeding is severe. Abbas p.488

M4-A3 Type A **Key: A**

A 4-year-old boy has a Staphylococcus aureus liver abscess. Last year he had invasive pulmonary aspergillosis. A lymph node biopsy shows granulomas. His maternal uncle died in childhood of a severe infection. Which molecular defect is most likely?

A ✔ Mutation in the gp91 subunit of cytochrome b558
About two-thirds of CGD is X-linked recessive, due to gp91 mutations. Catalase-producing Staphylococcus, Aspergillus and granulomas are typical; the maternal-uncle history fits X-linked inheritance. Abbas p.486-487

B Mutation in the p47 cytosolic oxidase subunit
Mutations in other oxidase subunits give autosomal recessive CGD, often milder with later onset. The maternal-uncle history favours the X-linked form. Janeway p.82

C Myeloperoxidase deficiency
MPO deficiency gives a CGD-like but less severe phenotype; it does not fit X-linked inheritance with life-threatening infections. Janeway p.536

D Mutation in the CD18 gene
LAD-1 is autosomal recessive and causes absent pus and leukocytosis, not granulomas with intact recruitment. Abbas p.488

E Mutation in IFN-γ receptor 1
IFN-γ receptor defects cause MSMD, with mycobacterial and Salmonella infections rather than Staphylococcus and Aspergillus. Abbas p.498

M4-A4 Type A **Key: B**

A 35-year-old woman has occasional bacterial infections that respond well to treatment. Laboratory studies show that her neutrophils produce normal amounts of superoxide and hydrogen peroxide. However, her neutrophils cannot convert hydrogen peroxide into hypochlorous acid. Which enzyme is deficient?

A Superoxide dismutase
SOD converts superoxide to hydrogen peroxide; her H_2O_2 production is normal. Abbas p.96

B ✔ Myeloperoxidase
MPO uses H_2O_2 to convert halide ions into hypohalous acids. MPO deficiency gives a CGD-like but milder phenotype. Abbas p.96; Janeway p.536

C NADPH oxidase (gp91 subunit)
NADPH oxidase makes superoxide; her superoxide production is normal, so CGD is excluded. Abbas p.96

D Inducible nitric oxide synthase
iNOS converts arginine to citrulline and NO; it does not make hypochlorous acid. Abbas p.96

E Neutrophil elastase
Elastase is a serine protease; ELA2 mutations cause severe congenital or cyclic neutropenia. Abbas p.96; Janeway p.534

M4-A5 Type A **Key: D**

A 3-year-old boy has recurrent pyogenic infections, silvery hair and partial oculocutaneous albinism. His blood film shows giant granules in neutrophils and lymphocytes. He now has fever, hepatosplenomegaly and hemophagocytosis. Which protein is most likely defective?

A GDP-fucose transporter
This causes LAD-2 with absent rolling and developmental delay, not giant lysosomes or albinism. Abbas p.488

B KINDLIN-3
This causes LAD-3 with integrin activation failure and bleeding, not albinism. Abbas p.488

C Neutrophil elastase (ELA2)
ELA2 mutations cause severe congenital or cyclic neutropenia, not giant granules or albinism. Janeway p.534

D ✔ LYST (lysosomal trafficking regulator)
Chédiak–Higashi syndrome: LYST mutation, giant lysosomes, partial albinism, silver hair. Without transplantation, HLH usually develops. Abbas p.488; Janeway p.540

E HAX1
HAX1 deficiency causes autosomal recessive severe congenital neutropenia, not CHS. Janeway p.534

M4-A6 Type A — **Key: C**

A 6-year-old boy has episodes of fever and bacterial infection about every 3 weeks. Serial blood counts show neutrophil numbers that swing from near normal to almost undetectable on a roughly 21-day cycle. Which gene is most likely mutated?

A HAX1
HAX1 deficiency causes autosomal recessive SCN with persistently low counts, not a cycle. Janeway p.534

B CD18
CD18 mutation causes LAD-1, typically with leukocytosis, not cyclic neutropenia. Abbas p.488

C ✔ ELA2 (neutrophil elastase)
Cyclic neutropenia has a cycle of about 21 days. Some ELA2 mutations cause cyclic neutropenia; ELA2 is also the most common cause of SCN. Janeway p.533-534

D LYST
LYST mutation causes CHS with giant lysosomes and albinism. Abbas p.488

E IL-12Rβ1
IL-12Rβ1 mutations cause MSMD with mycobacterial infections, not cyclic neutropenia. Abbas p.498

M4-A7 Type A — **Key: A**

A 1-year-old girl has recurrent bacterial infections, a high blood leukocyte count, severe developmental delay and the Bombay blood group phenotype. Which step of neutrophil recruitment is primarily blocked?

A ✔ Selectin-mediated rolling on endothelium
LAD-2: a defective GDP-fucose transporter prevents synthesis of sialyl Lewis X, the selectin ligand, so leukocytes do not roll. Fucosylation failure also causes the Bombay phenotype. Abbas p.488

B Chemokine-induced integrin activation
This step fails in LAD-3 (KINDLIN-3), which also causes bleeding. Abbas p.488

C Integrin binding to ICAM-1 for firm arrest
This step fails in LAD-1 (CD18), without developmental delay or Bombay phenotype. Abbas p.488

D Paracellular transmigration via CD31
CD31 and VE-cadherin act at transmigration; no LAD type in the sources targets this step. Abbas p.50

E Phagosome fusion with lysosomes
This is a post-ingestion step that fails in CHS, not a recruitment step. Janeway p.536

M4-K1 KPrim — **Key: + / – / + / –**

Regarding leukocyte recruitment from blood into infected tissue, which of the following statements is/are correct?

A (+) IL-1 and TNF induce E-selectin on endothelial cells within 1 to 2 hours.
Correct. E-selectin is synthesized and expressed within 1 to 2 hours after IL-1 and TNF. Abbas p.45

B (–) LFA-1 on neutrophils binds mainly VCAM-1 on endothelium.
Incorrect. LFA-1 binds ICAM-1; VCAM-1 is the ligand for VLA-4. Abbas p.45

C (+) Chemokine receptor signals raise integrin affinity by inside-out signalling.
Correct. Chemokine-induced intracellular signals change the integrin to a high-affinity form; this is inside-out signalling. Abbas p.46

D (–) Neutrophils are recruited mainly through CCR2 binding to CCL2.
Incorrect. Neutrophils use CXCR1 and CXCR2 (CXCL1, CXCL8); CCR2 and CCL2 recruit classical monocytes. Abbas p.50

M4-K2 KPrim — **Key: + / – / + / +**

Regarding microbial killing by neutrophils, which of the following statements is/are correct?

A (+) Phagocyte oxidase reduces molecular oxygen to superoxide, with NADPH as cofactor.
Correct. This is the first reaction of the respiratory burst. Abbas p.96

B (–) Myeloperoxidase converts superoxide into hydrogen peroxide.
Incorrect. Superoxide is dismutated to H_2O_2 (by SOD); MPO then uses H_2O_2 to make hypohalous acids. Abbas p.96; Janeway p.81

C (+) Catalase made by Staphylococcus destroys hydrogen peroxide, which helps explain this organism's importance in CGD.
Correct. Catalase destroys the H_2O_2 that host cells could make from residual superoxide. Abbas p.487

D (+) NET formation requires citrullination of histones by PAD4.
Correct. NET formation needs PAD4, elastase, myeloperoxidase and phagocyte oxidase. Abbas p.96

M4-K3 KPrim **Key: + / + / – / –**

Regarding Mendelian susceptibility to mycobacterial disease (MSMD), which of the following statements is/are correct?

A (+) Live BCG vaccine can cause disseminated infection in affected patients.
Correct. In MSMD, BCG vaccine can cause disseminated mycobacterial infection. Abbas p.498

B (+) IL-12p40 deficiency removes both IL-12 and IL-23 activity.
Correct. p40 is shared by IL-12 and IL-23. Janeway p.538

C (–) Total absence of the IFN-γ receptor causes exaggerated granuloma formation.
Incorrect. Total absence of IFN-γ receptor expression markedly reduces granuloma formation. Janeway p.538

D (–) The typical severe infections are invasive Aspergillus and catalase-positive staphylococci.
Incorrect. That pattern fits CGD. MSMD involves weakly virulent mycobacteria and other intracellular pathogens such as Salmonella. Abbas p.487, 498

M5-A1 Type A **Key: C**

A 19-year-old male student has his second episode of meningococcal meningitis in two years. He has had no other serious infections. Complement C3 and C4 levels are normal. A defect in which complement function best explains this pattern?

A Opsonization of bacteria by C3b and iC3b
Loss of C3b opsonization (e.g., C3 deficiency) causes infections with a wide range of extracellular bacteria, not Neisseria alone; C3b and iC3b opsonize via CR1, CR3 and CR4. Janeway p.64; Abbas p.313

B Mast cell activation by the anaphylatoxins C3a and C5a
Anaphylatoxins drive inflammation and mast cell degranulation; their loss does not explain selective Neisseria susceptibility. Abbas p.313

C ✔ Osmotic lysis by the membrane attack complex (C5b-9)
MAC lysis is critical only against a few thin-walled pathogens such as Neisseria, and C5-C9 deficiency is linked to Neisseria infections only. Abbas p.314; Janeway p.66; Abbas p.307

D Transport of immune complexes by CR1 on erythrocytes
CR1 on erythrocytes carries C3b/C4b-coated immune complexes to liver and spleen; this does not explain recurrent Neisseria. Abbas p.307

E B-cell co-stimulation by C3d binding to CR2 (CD21)
C3d-CR2 enhances B-cell activation; it is not the key defense against Neisseria. Abbas p.314; Abbas p.308

M5-A2 Type A **Key: E**

A 45-year-old woman develops flushing, hypotension and wheeze during massive complement activation. Specific IgE to all tested triggers is negative. Which complement fragment is the most potent direct inducer of mast cell degranulation in this setting?

A C3b
C3b is the large opsonin fragment that binds covalently to surfaces; it is not an anaphylatoxin. Abbas p.301; Abbas p.313

B C4a
C4a is an anaphylatoxin, but it is about 2500-fold less potent than C5a. Abbas p.314

C C3a
C3a degranulates mast cells, but it is about 20-fold less potent than C5a. Abbas p.314; Abbas p.470

D C5b-9
C5b-9 is the MAC pore that lyses cells; it is not the anaphylatoxin that degranulates mast cells. Abbas p.307

E ✔ C5a
C5a is the most potent mediator of mast cell degranulation, and C3a/C5a degranulate mast cells without IgE. Large systemic amounts cause a shock-like syndrome resembling systemic allergy. Abbas p.314; Abbas p.470; Janeway p.64

M5-A3 Type A **Key: A**

A 32-year-old man has episodes of intravascular hemolysis. Flow cytometry shows that his erythrocytes lack GPI-anchored membrane proteins. Which pair of complement-control functions is missing on his red cells?

A ✔ Dissociation of C3 convertases (DAF) and blocking of C9 binding to C5b-8 (CD59)
DAF and CD59 are GPI-linked; DAF dissociates both C3 convertases and CD59 blocks C9 addition to C5b-8. Loss of GPI anchors removes both (PNH). Abbas p.310; Abbas p.312; Abbas p.311

B Dissociation of C1r2s2 from C1q (C1-INH) and inhibition of MASP2
C1-INH is a plasma serpin, not a GPI-anchored membrane protein. Abbas p.311

C Displacement of C2 from C4b in plasma (C4BP) and factor I cofactor activity
C4BP is a plasma regulator, so GPI-anchor loss does not affect it. Abbas p.310

D Binding of S protein (vitronectin) to soluble C5b-7 complexes
S protein is a plasma protein that binds soluble C5b-7; it is not membrane-anchored. Abbas p.312; Janeway p.70

E Factor H binding to sialic acid and displacement of Bb from C3b
Factor H is a plasma regulator; it does not depend on a GPI anchor. Abbas p.310; Abbas p.311

M5-A4 Type A **Key: D**

A 4-year-old child has microangiopathic hemolytic anemia, thrombocytopenia and acute kidney injury. Genetic testing shows a heterozygous mutation in the factor H gene. Which normal function of factor H is most likely reduced?

A Direct proteolytic cleavage of C3b into iC3b without any cofactor
Factor I, not factor H, is the protease; factor H is one of its cofactors. Abbas p.310; Janeway p.68

B Inhibition of C1r, C1s and MASP2 by covalent binding
This is the function of C1-INH, a serpin. Abbas p.311

C Blocking of C9 polymerization on the C5b-8 complex
This is the function of CD59 (protectin). Abbas p.312; Janeway p.70

D ✔ Binding of C3b on host cells to displace Bb and to act as factor I cofactor
Factor H binds C3b, displaces Bb and is a cofactor for factor I. It is the gene most often mutated in complement-mediated hemolytic-uremic syndrome. Abbas p.310; Abbas p.315; Abbas p.311

E Stabilization of the C3bBb convertase on microbial surfaces
Properdin stabilizes C3bBb; factor H does the opposite. Abbas p.302

M5-A5 Type A **Key: B**

A 6-year-old boy has lobar pneumonia caused by Streptococcus pneumoniae. His serum C-reactive protein (CRP) is very high. CRP bound to the pneumococcal surface activates complement by binding which molecule?

A Mannose-binding lectin (MBL)
MBL is itself a recognition molecule of the lectin pathway; it binds mannose residues, not CRP. Abbas p.306

B ✔ C1q
C1q binds pentraxins such as CRP and serum amyloid P, so CRP activates the classical pathway. Abbas p.305

C Factor B
Factor B binds surface C3b in the alternative pathway; it does not bind CRP. Abbas p.301

D Properdin
Properdin stabilizes the alternative pathway convertase C3bBb. Abbas p.302

E L-ficolin
Ficolins are lectin-pathway sensors that bind N-acetylglucosamine-containing glycans. Abbas p.306

M5-K1 KPrim **Key: + / – / + / –**

Complement convertases. Which of the following statements is/are correct?

A (+) The classical and lectin pathways form the same C3 convertase, C4b2a.
C4b2a is the classical C3 convertase, and the lectin pathway forms the same C4b2a. Abbas p.305; Janeway p.54

B (–) Factor D circulates as an inactive zymogen and is activated by C3b.
Factor D is the only activating complement protease that circulates as an active enzyme, not a zymogen. Janeway p.61

C (+) The alternative pathway C5 convertase contains one Bb and two C3b molecules.
Abbas describes the alternative C5 convertase as one Bb with two C3b; Janeway writes C3b2Bb. Abbas p.303; Janeway p.62

D (–) C5b binds covalently to the target membrane through a thioester bond.
Unlike C3b and C4b, C5b does not bind covalently to the cell surface. Janeway p.62

M5-K2 KPrim **Key: + / + / – / –**

Complement receptors and regulators. Which of the following statements is/are correct?

A (+) C1-INH inhibits C1r, C1s and MASP2, and it also inhibits kallikrein and factor XII.
C1-INH is a serpin that inactivates C1r/C1s and MASP2, and also kallikrein and coagulation factor XII. Abbas p.311

B (+) CR2 (CD21) on B cells binds C3d and forms a co-receptor complex with CD19 and CD81.
CR2 binds C3d, C3dg and iC3b; on B cells it forms a complex with CD19 and CD81, and C3d-CR2 facilitates B-cell activation. Abbas p.308; Abbas p.314

C (–) CD59 protects host cells by dissociating the C3 convertase C3bBb.
CD59 blocks C9 addition to C5b-8; DAF dissociates the C3 convertases. Abbas p.312; Abbas p.310

D (–) Factor H regulates both pathways because it displaces C2a from C4b2a.
Factor H blocks only Bb binding to C3b, so it regulates the alternative but not the classical pathway; C4BP and DAF displace C2/C2a. Abbas p.311; Abbas p.310

M6-A1 Type A **Key: C**

A 19-year-old man has his second episode of meningococcal meningitis in 3 years. His maternal uncle died of meningococcal sepsis at age 20. His mother and sisters are healthy. Which inherited complement deficiency best explains this pattern?

A C2 deficiency
C2 deficiency is linked mainly to SLE-like immune-complex disease, and is not X-linked. Abbas p.315

B C8 deficiency
Terminal component deficiency also causes Neisseria infection, but the X-linked pattern (affected males through the maternal line) fits properdin better. Abbas p.315

C ✔ Properdin deficiency
Properdin deficiency is X-linked recessive, and properdin-deficient patients are particularly susceptible to Neisseria meningitidis. Abbas p.315; Janeway p.59

D MBL deficiency
MBL deficiency gives a mild immunodeficiency with bacterial infections in early childhood, not recurrent meningococcal disease in adults. Janeway p.533

E C1q deficiency
C1q deficiency causes SLE in more than 50% of patients; Neisseria susceptibility is the hallmark of alternative and terminal defects. Abbas p.315

M6-A2 Type A **Key: A**

A 24-year-old woman has a photosensitive rash, arthritis and positive antinuclear antibodies. She also had two episodes of bacterial pneumonia. Genetic testing finds a homozygous deficiency of a single complement component. This component deficiency is the most common in humans. Which component is deficient?

A ✔ C2
C2 deficiency is the most common human complement deficiency. More than 50% of patients with C2 deficiency develop SLE, and some have more infections. Abbas p.315

B C3
C3 deficiency mainly causes frequent, serious, possibly fatal pyogenic infections; it is not the most common deficiency. Abbas p.315

C C6
Terminal component deficiency causes Neisseria infection as the only consistent problem, not SLE. Abbas p.315

D Factor I
Factor I deficiency depletes C3 and causes pyogenic infections; it is not the most common deficiency. Abbas p.315

E Properdin
Properdin deficiency causes meningococcal infection, not an SLE-like illness. Abbas p.315

M6-A3 Type A **Key: D**

A 7-year-old boy has had recurrent pneumonia and one episode of bacteraemia with pyogenic bacteria. Serum C3 is markedly low. Genetic testing shows a homozygous factor I mutation; the C3 gene is normal. What best explains the low C3?

A Reduced hepatic synthesis of C3
The C3 gene is normal; the sources explain the low C3 by consumption, not by reduced synthesis. Abbas p.315

B An autoantibody that destroys C3
The autoantibody C3NeF stabilizes C3bBb; it does not destroy C3, and this boy has a genetic defect. Abbas p.315

C Loss of GPI-anchored regulators on blood cells
Loss of GPI-anchored DAF and CD59 causes PNH with haemolysis, not C3 depletion. Abbas p.315

D ✔ Uncontrolled fluid-phase C3 convertase that consumes C3
Without factor I, fluid-phase C3 convertase forms without control and depletes C3; the result is a de facto C3 deficiency with pyogenic infections. Abbas p.315; Janeway p.533

E Failure of C1-INH to block C1r and C1s
Loss of C1-INH control causes consumption of C4 and C2 and HAE; C3 is expected to be normal in HAE. Janeway p.67; HAE-GL p.6

M6-A4 Type A **Key: E**

A 16-year-old girl has recurrent painful abdominal attacks and swelling of the hands since age 11. Her father has similar episodes. Antihistamines did not help. She has no wheals. Which laboratory profile is most likely, if she has the most common form of hereditary angioedema?

A C1-INH function normal, C1-INH protein normal, C4 normal
Normal C1-INH level and function fit HAE with normal C1-INH, which is very rare. HAE-GL p.6

B C1-INH function low, C1-INH protein normal or high, C4 low
This is HAE-2 (dysfunctional C1-INH), the less common form. HAE-GL p.5

C C1-INH normal, C3 low, CH50 low
C3 and CH50 are expected to be normal in HAE and are usually not helpful. HAE-GL p.6-7

D C1-INH function low, C1-INH protein low, C1q low
Low C1q points to acquired C1-INH deficiency; C1q is nearly always normal in HAE. HAE-GL p.7

E ✔ C1-INH function low, C1-INH protein low, C4 low
HAE-1 is about 85% of patients and has low C1-INH concentration and function; C4 is usually low. HAE-GL p.6

M6-A5 Type A **Key: B**

A 63-year-old man has had four episodes of facial and tongue swelling in 6 months. He has no family history of angioedema and no wheals. He takes no ACE inhibitor. C1-INH function, C1-INH protein and C4 are all low on two occasions. Which test is the most useful next step?

A Serum tryptase
The low C1-INH profile already points to a bradykinin-mediated, C1-INH-related angioedema; the guideline next step is C1q. HAE-GL p.7

B ✔ Serum C1q
C1q is recommended in new-onset angioedema after age 40 without family history; C1q is low in 75% of AAE-C1-INH and nearly always normal in HAE. HAE-GL p.7

C SERPING1 gene sequencing
Sequencing can support HAE diagnosis, but late onset without family history first needs C1q to look for acquired deficiency. HAE-GL p.7

D C3 and CH50
C3 and CH50 are expected to be normal in HAE and are usually not helpful. HAE-GL p.6-7

E F12 gene analysis
F12 mutations cause HAE with normal C1-INH; this patient has low C1-INH. HAE-GL p.5-6

M6-A6 Type A **Key: C**

A 29-year-old woman has recurrent lip and facial swellings lasting 2-3 days, without wheals. They became more frequent after she started a combined oral contraceptive. Her father has similar swellings. High-dose antihistamines did not help. C1-INH concentration and C1-INH function are normal on repeated testing. What is the most likely diagnosis?

A HAE-2 (dysfunctional C1-INH)
HAE-2 has low functional C1-INH; her function is normal. HAE-GL p.5

B Acquired angioedema with C1-INH deficiency
AAE-C1-INH has low C1-INH function and protein, later onset and often low C1q. HAE-GL p.7

C ✔ Hereditary angioedema with normal C1-INH
HAE nC1-INH (for example HAE-FXII) resembles HAE-1/2 clinically but C1-INH level and function are normal; the family history supports a hereditary form. HAE-GL p.5-6

D ACE inhibitor-induced angioedema
ACEI-AE is bradykinin-mediated with normal C1-INH, but she takes no ACE inhibitor and has a family history. HAE-GL p.6

E Mast cell-mediated angioedema of chronic spontaneous urticaria
Absence of wheals does not exclude CSU, but the family history and failure of antihistamines favour a hereditary bradykinin form. HAE-GL p.7

M6-A7 Type A **Key: D**

A 35-year-old man with known HAE-1 comes to the emergency department 1 hour after onset of tongue swelling and a hoarse voice. He has no stridor yet. According to the 2017 WAO/EAACI guideline, what is the best management?

A Intramuscular adrenaline, IV antihistamine and IV glucocorticoid, then observe
Failure to respond to antihistamines, glucocorticoids or epinephrine is typical of HAE; the swelling is bradykinin-mediated. HAE-GL p.6

B High-dose oral tranexamic acid and admission for observation
Antifibrinolytics show no or only minimal effect for on-demand treatment and are not advised. HAE-GL p.8

C Start danazol and review the response after 24 hours
Androgens are not advised for on-demand treatment of attacks. HAE-GL p.8

D ✔ Give C1-INH concentrate or icatibant at once and prepare early airway intervention
Treat any attack that affects the upper airway, as early as possible, with C1-INH, ecallantide or icatibant; consider intubation or surgical airway early if edema progresses. HAE-GL p.7-8

E Observe without drugs because most attacks resolve without treatment
The guideline recommends treating any attack that affects or may affect the upper airway; laryngeal attacks are emergencies. HAE-GL p.7-8

M6-A8 Type A **Key: E**

A 31-year-old woman with HAE-1 is 16 weeks pregnant. Her attacks have increased to three per month, with two abdominal attacks. She asks for long-term prophylaxis. According to the 2017 WAO/EAACI guideline, which option is preferred?

A Danazol at the lowest effective dose

Androgens are contraindicated in pregnancy because they cross the placenta. HAE-GL p.13

B Icatibant twice weekly

Icatibant is an on-demand B2 receptor antagonist, not a long-term prophylactic drug; C1-INH is preferred in pregnancy. HAE-GL p.8, p.13

C Ecallantide every 3 days

Ecallantide is an on-demand kallikrein inhibitor licensed only in the USA and needs a healthcare professional because of anaphylaxis risk. HAE-GL p.8

D Tranexamic acid as first choice

Antifibrinolytics are not recommended for long-term prophylaxis; in pregnancy they may be considered only if C1-INH concentrate is unavailable. HAE-GL p.10, p.13

E ✔ Plasma-derived C1-INH concentrate

C1-INH concentrate is the recommended first-line therapy in pregnancy and is safe and effective when long-term prophylaxis is needed; it is also first-line long-term prophylaxis in general. HAE-GL p.9-10, p.12-13

M6-K1 KPrim **Key: + / – / + / –**

Complement deficiencies and their phenotypes. Which of the following statements is/are correct?

A (+) Deficiency of C5, C6, C7, C8 or C9 predisposes mainly to disseminated Neisseria infection.

Neisseria infection is the only consistent clinical problem in terminal component deficiency, because MAC lysis matters for these thin-walled bacteria. Abbas p.315

B (–) Properdin deficiency is inherited as an autosomal dominant trait.

Properdin deficiency is X-linked recessive. Abbas p.315

C (+) More than half of patients with C1q deficiency develop SLE.

More than 50% of patients with C1q, C2 or C4 deficiency develop SLE. Abbas p.315

D (–) MBL deficiency typically causes severe lifelong pyogenic infections similar to C3 deficiency.

MBL deficiency is common and may cause only a mild immunodeficiency with bacterial infections in early childhood. Janeway p.533

M6-K2 KPrim **Key: + / + / – / +**

Management of HAE-1/2 according to the WAO/EAACI guideline, 2017 revision. Which of the following statements is/are correct?

A (+) Icatibant is a competitive antagonist of the bradykinin B2 receptor.

Icatibant is a synthetic peptide that selectively and competitively blocks the B2 receptor. HAE-GL p.8

B (+) C1-INH concentrate is the preferred short-term prophylaxis before a dental extraction.

Short-term prophylaxis is recommended before procedures that can induce an attack, preferably with C1-INH concentrate close to the procedure. HAE-GL p.9

C (–) Attenuated androgens are the recommended first-line long-term prophylaxis.

C1-INH is first-line long-term prophylaxis (Rec. 10); androgens are second line (Rec. 11). HAE-GL p.9-10

D (+) Every patient should have on-demand medication for two attacks and carry it at all times.

This is Recommendation 6. HAE-GL p.8

M6-K3 KPrim

Key: – / + / + / –

Diseases caused by defective complement regulators. Which of the following statements is/are correct?

A (–) PNH results from a germline PIG-A mutation inherited from a parent.
The PNH mutation is not inherited; it is an acquired somatic mutation in haematopoietic stem cells. Abbas p.312

B (+) Mutations in factor H, MCP or factor I predispose to atypical haemolytic uraemic syndrome.
Abbas names factor H as the most common mutated gene in aHUS, with MCP also mutated in many patients; Janeway adds factor I, and heterozygous mutations can predispose. Abbas p.315; Janeway p.70

C (+) Factor I deficiency causes secondary C3 depletion and pyogenic infections.
Uncontrolled fluid-phase C3 convertase depletes C3, which increases pyogenic infections. Abbas p.315

D (–) Factor H mainly regulates the classical pathway C3 convertase.
Factor H blocks only the binding of Bb to C3b, so it regulates the alternative pathway but not the classical pathway. Abbas p.311

SCT-S1 SCT

A 38-year-old woman has had recurrent swellings of the lips and hands for several years and two episodes of severe abdominal pain. She has no wheals. Daily antihistamines did not prevent the swellings. You consider the causes of recurrent angioedema. (Provisional panel credits.)

Hypothesis: HAE-1/2 (C1-INH deficiency or dysfunction) · **New info:** C1-INH function is low and C4 is low, confirmed on a repeat sample.
Credits: -2: 0, -1: 0, 0: 0, +1: 0.5, +2: 1
Low functional C1-INH is the shared finding of HAE-1 and HAE-2, and C4 is usually low in HAE-1/2. Recurrent swelling, abdominal attacks, no wheals and no antihistamine response also fit HAE-1/2. HAE-GL p.5-6

Hypothesis: Acquired angioedema with low C1-INH (AAE-C1-INH) · **New info:** C1q is normal.
Credits: -2: 0.5, -1: 1, 0: 0.2, +1: 0, +2: 0
C1q is low in 75% of AAE-C1-INH and nearly always normal in HAE. A normal C1q makes AAE-C1-INH less likely but does not exclude it (25% normal; also normal with anabolic androgens). HAE-GL p.7

Hypothesis: HAE with normal C1-INH (HAE nC1-INH) · **New info:** C1-INH function, C1-INH protein and C4 are normal on two occasions, and her mother has the same swellings.
Credits: -2: 0, -1: 0, 0: 0.2, +1: 1, +2: 0.7
HAE nC1-INH has normal C1-INH and clinically resembles HAE-1/2; a family history supports a hereditary form. It is very rare and most patients have no mutation found, so a strong shift is debatable. HAE-GL p.6

SCT-S2 SCT

A 6-year-old boy has had recurrent bacterial infections since infancy, including skin abscesses and two pneumonias. Growth is normal. You consider a phagocyte defect or a complement deficiency. (Provisional panel credits.)

Hypothesis: Chronic granulomatous disease (CGD) · **New info:** He now has a liver abscess with Staphylococcus aureus and a lung infection with Aspergillus.
Credits: -2: 0, -1: 0, 0: 0, +1: 0.5, +2: 1
Catalase-producing organisms such as Staphylococcus are troublesome in CGD, and invasive Aspergillus is the leading cause of death in CGD. Phagocyte defects typically involve skin, liver and respiratory tract. Abbas p.486-487

Hypothesis: Terminal complement (C5-C9) deficiency · **New info:** He has had two episodes of disseminated Neisseria meningitidis infection.
Credits: -2: 0, -1: 0, 0: 0, +1: 0.6, +2: 1
In C5-C9 deficiency the only consistent clinical problem is disseminated Neisseria infection, because MAC lysis is important against these thin-walled bacteria. Properdin and factor D defects give a similar picture, so +1 earns partial credit. Abbas p.315

Hypothesis: Leukocyte adhesion deficiency type 1 (LAD-1) · **New info:** The skin abscesses contained abundant pus.
Credits: -2: 1, -1: 0.6, 0: 0, +1: 0, +2: 0
LAD causes failure of neutrophil recruitment and inability to make pus; infections show no pus. Abundant pus argues strongly against LAD-1. Abbas p.488

SCT-S3 SCT

A 9-year-old girl has had recurrent episodes of fever with an urticaria-like rash, joint pain and red eyes since early childhood. Episodes often follow exposure to cold. Her mother had similar episodes as a child. (Provisional panel credits.)

Hypothesis: Cryopyrin-associated periodic syndrome (CAPS, NLRP3 gain of function) · **New info:** Her mother has progressive hearing loss.

Credits: -2: 0, -1: 0, 0: 0.2, +1: 0.6, +2: 1

Muckle-Wells syndrome and FCAS are dominantly inherited NLRP3 disorders with fever, urticarial rash, joint pain and conjunctivitis; Muckle-Wells also causes progressive deafness. Janeway p.537

Hypothesis: Familial Mediterranean fever (FMF) · **New info:** Genetic testing finds a gain-of-function NLRP3 mutation.

Credits: -2: 0.7, -1: 1, 0: 0, +1: 0, +2: 0

FMF is caused by MEFV (pyrin) mutations, whereas CAPS is caused by gain-of-function NLRP3 mutations. A disease-causing NLRP3 mutation explains the picture and makes FMF much less likely. Abbas p.74-75

Hypothesis: CAPS (NLRP3 gain of function) · **New info:** The attacks respond dramatically to anakinra, an IL-1 receptor antagonist.

Credits: -2: 0, -1: 0, 0: 0.3, +1: 1, +2: 0.5

CAPS is driven by excess IL-1 and responds to IL-1 antagonists such as anakinra. A response supports CAPS but is not specific, because IL-1 blockade also helps other IL-1-driven conditions such as severe gout. Abbas p.74; Janeway p.537